

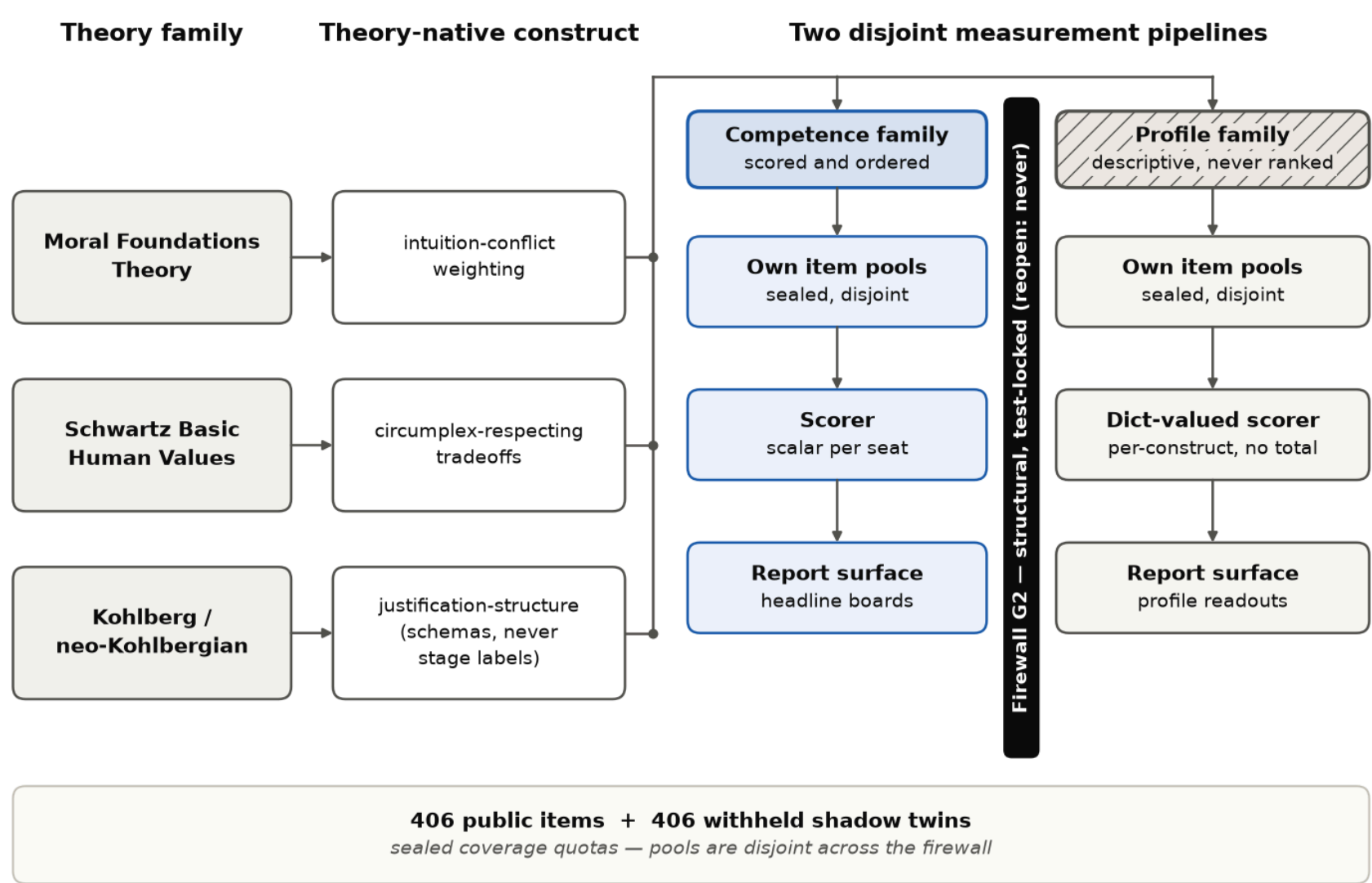
# Calibrating the Judge Before It Grades

## Bidirectional Alignment as a Preregistered Measurement Gate

Anonymous Author(s) · BiAlign 2026 workshop submission (preregistered report) · v2

**Gate-M status: PENDING.** Descriptive, exploratory, pre-Gate-M throughout — every score on this poster is the output of an unvalidated judge or coder. Validity is Gate-M-only, and Gate M has not run.

No existing benchmark operationalizes all three moral-psychology schools — Moral Foundations Theory, Schwartz Basic Human Values, and Kohlberg and neo-Kohlbergian moral development — as separate theory-native constructs, and none applies the competence-versus-default-profile firewall to them. The contribution is that conjunction, under measurement-disciplined conditions.



Three schools, three constructs, across Firewall G2.

Sealed: **406 public items paired one-to-one with 406 withheld shadow twins — 812 item surfaces in all — scored against 352 sealed rubrics.**

- Gate M, direction one:** trained humans calibrate the machine judge before it grades any headline.
- Direction two:** the expert–expert ceiling bounds what the machine may claim.

Sealed floor, agreement =  $1 - \text{mean}|\Delta|$  on the 0-to-1 scale — the panel passes only if both hold:

- $\text{agreement}(\text{judge}, \text{expert}) \geq \text{agreement}(\text{expert}, \text{expert}) - 0.10$
- $|\text{bias}| \leq 0.15$

**No code path lets a judge that misses the floor grade a headline;** a partial return routes to demotion, **never to a pass.**

Gate-M constants are sealed preregistered design values (PM1-C004) — not measured quantities. No expert ratings exist as of this writing; no expert-validated judge-performance claim is made anywhere (PM1-C093).

B2-01

### Competence family — dual-judge consistency, never pooled

**Thirteen seats across four disjoint substrates**, never merged into one board. Every seat receives the same geometry — **120 subject cells**, of which **102 are judged** — so a dual-judged seat yields **204 verdicts** and a single-judged seat **102**, under a sealed leave-one-out firewall. Subject-cell denominators are 360 / 360 / 240 / 600, all census-clean on their own declared denominators, with **1 disclosed no-verdict**.

**Consistency is not validity:** agreement between two unvalidated judges is no evidence that either is correct, and more cells never convert one into the other. The two dual-judged substrates are **never pooled** — different items, seats, dates, sizes — and "within 0.1" uses **one named convention for both** (the epsilon rule, 1e-9 tolerance).

| statistic                            | first dual-judged substrate<br>(306 paired cells, 3 seats) | second dual-judged<br>substrate (510 paired cells,<br>5 seats) |
|--------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|
| mean $ \Delta $                      | 0.1618                                                     | 0.1476                                                         |
| median $ \Delta $                    | 0.0500                                                     | 0.0000                                                         |
| exact agreement                      | 0.471                                                      | 0.537                                                          |
| within-0.1 (epsilon convention)      | 0.670                                                      | 0.749                                                          |
| within-0.25                          | 0.850                                                      | 0.839                                                          |
| signed mean $\Delta$ (leg 2 – leg 1) | +0.0403                                                    | +0.0599                                                        |
| max $ \Delta $                       | 1.00                                                       | 1.00                                                           |
| no-verdicts, either leg, final       | 0                                                          | 0                                                              |

A judge-pair property, never a seat property. Both judges unvalidated pre-Gate-M; the two legs are **never averaged into a single number**. Rows and columns are in declared roster order, never a score order. Establishes no ranking (PM1-C090), tier (PM1-C091), hypothesis verdict (PM1-C092) or judge validity (PM1-C093).

The offset runs the same **direction** on both substrates and is larger on the second. The second surface's reproducibility reference was **seeded by its own first run**, so recomputation shows reproducibility of the computation, never validity of the judgments. This is the quantity a single-judge design cannot produce at all — the measurement argument for the gate, not a substitute for it. The board is **not a representative sample** of any model population; "frontier" names a vendor's flagship line, never a measured competence tier.

B2-02, B2-03, B2-04, B2-09, B2-10

### Profile family — coder-agreement channels, never averaged

Coverage is **complete sixteen-of-sixteen** on the sealed roster — **5 + 8 + 3 = 16** across three provenance tiers, disjoint by construction and enforced as disjoint at import. Every figure here is an **inter-coder agreement rate between two independent cross-vendor coders**: how often two coders assigned the same code to the same cell. It is a property of the measurement instrument, not of any model. High agreement means the instrument is reproducible on that channel; it does not mean the code is correct, since two coders can agree and both be wrong.

Profiles are **never ranked**, and not merely pre-Gate-M: the three profile constructs have **no better/worse direction**, so passing the gate would change what can be believed about the *coders* and would not create a direction on the *constructs*.

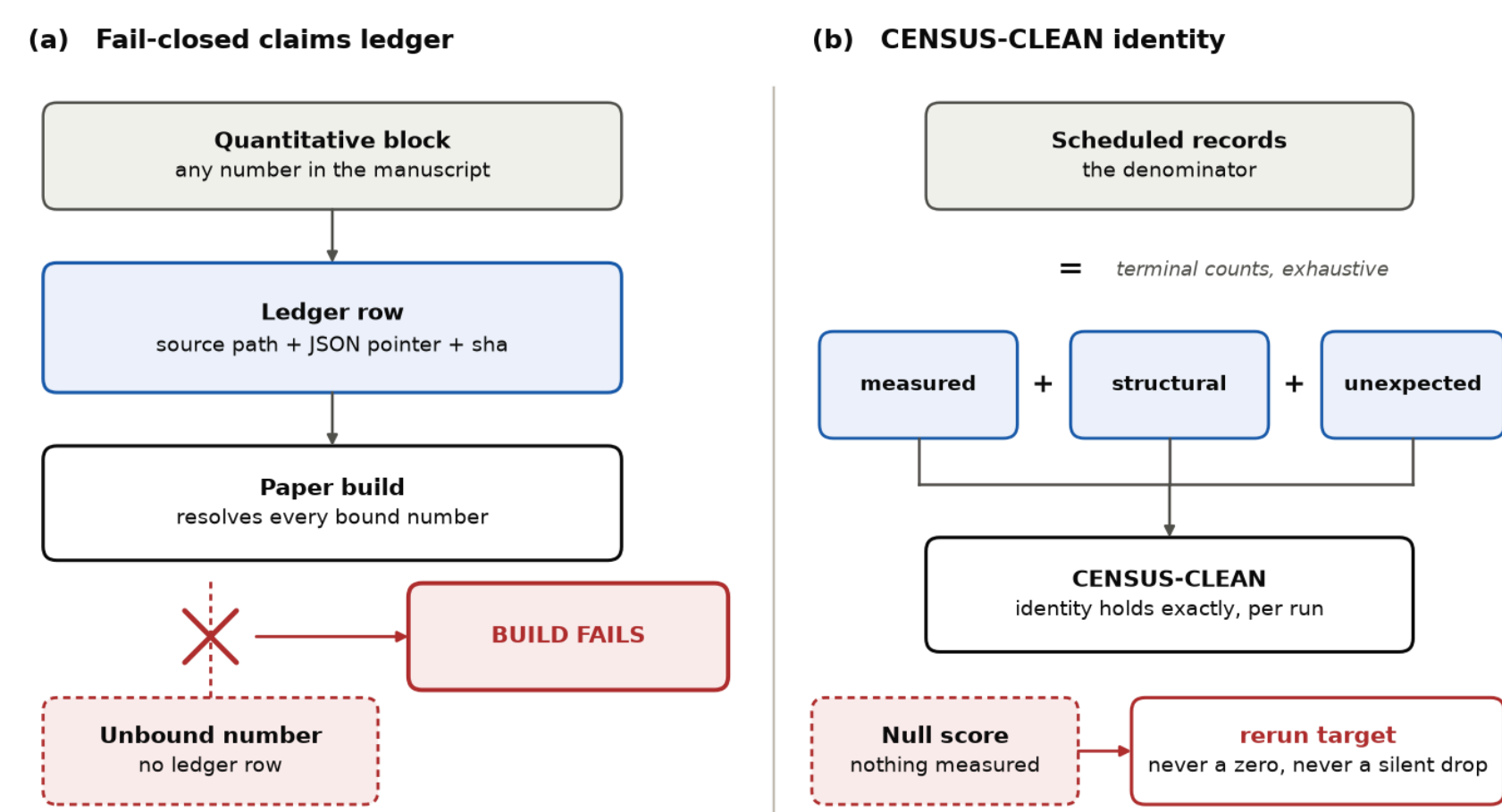
| construct               | channel          | pooled exact agreement | n   |
|-------------------------|------------------|------------------------|-----|
| intuition-weight        | care             | 0.785                  | 512 |
| intuition-weight        | equality         | 0.697                  | 512 |
| intuition-weight        | proportionality  | 0.738                  | 512 |
| intuition-weight        | loyalty          | 0.660                  | 512 |
| intuition-weight        | authority        | 0.689                  | 512 |
| intuition-weight        | purity           | 0.943                  | 512 |
| justification-structure | primary schema   | 0.827                  | 608 |
| justification-structure | secondary schema | 0.648                  | 608 |
| justification-structure | register         | 0.933                  | 608 |
| justification-structure | coded confidence | 0.538                  | 608 |

Descriptive, exploratory, pre-Gate-M; coders unvalidated. Agreement is a property of the coder pair, never of a seat, and no seat is named. Channels are reported separately and are **never averaged into one number**. No ordering, tier, or composite is expressed or implied by row order.

The finding is about **channels, not seats**; every ordering is over the coder pair's reproducibility on a channel, never over the constructs. The coder pair's least reproducible channel by a wide margin is coded confidence at **0.538** — an analysis leaning on it should pre-register a threshold or decline to use it. Both coder legs returned **210/210, census-clean**, at **\$0 cash**, on a scheduled census of **822 cells, 822 terminal, 813 measured**, with **9** in a disclosed mechanical-extraction gap outside every coder denominator. **Eleven** seats are extension-beside and **explicitly non-confirmatory**; provenance reports where a measurement came from, never how good a seat is. Profile ranking is prohibited outright and structurally (PM1-C095; reopen condition: never).

B2-11, B2-13, B2-14, B2-17, B2-23

### Governance as code



Fail-closed ledger, census identities, verdict ladder.

- Fail-closed ledger:** no unbound number ships.
- DECLINE** is preregistered — below a sealed floor, **the refusal is the reported result**.
- NOT\_EVALUABLE** by design, not a pass.
- Gate M: PENDING.** Machine side complete and census-clean, human side in flight, **no expert ratings exist**; every quantity is **quarantined-provisional**.

### Instrument vintage

Every score in this section was produced under the instrument's prior 2026-08-08 seal and the pre-revision judge prompt; the 2026-08-09 re-seal of record post-dates every reported measurement — it changed no counts, but it revised judge-prompt rendering and nineteen item surfaces, so no number here was produced under, or is comparable to output of, the current sealed instrument.

### Limitations

- No validated judge.** All scoring evidence now is **feasibility evidence**; the gate's failure is itself a reportable outcome.
- Rosters.** The two boards are stated separately and never compared (Firewall G2); neither is a **representative sample**, so ranking, tiers, hypothesis verdicts and generalization are prohibited.
- Scope.** These three schools measure resemblance to three specific, contested, culturally situated theories. They do **not** measure whether a model is moral, and no sentence in this program says otherwise.
- Interpretation, fixed in advance:** this evidence **supports the measurement architecture, not stable ordering** (PM1-C050).

**Evidence banner — verbatim, and it travels with every quantity on this poster.**

- All results in §8 are **descriptive, exploratory, pre-Gate-M** results on the program's measured boards, and they inherit the render law of the reports they bind to.
- Every score is the output of an **unvalidated** judge or coder, or an agreement between two unvalidated ones; **validity is Gate-M-only**, Gate M has not run, and agreement between two unvalidated instruments is never validity.
- Where a substrate is dual-judged there is **one table per judge**, and the two legs are **never averaged into a single number**; where a substrate is single-judged, its surface is one unvalidated judge's output alone.
- Substrates are **never merged**: coverage grew by adding substrates, not by growing one, and verdict denominators differ across them. Seat rows appear in their declared roster order, which is **never a score order**.
- Nothing here establishes a ranking (PM1-C090), a competence tier (PM1-C091), a hypothesis verdict (PM1-C092), judge validity (PM1-C093), or generalization beyond these seats (PM1-C094).

**Colophon.** Bound values displayed on this poster: B2-01, B2-02, B2-03, B2-04, B2-09, B2-10, B2-11, B2-13, B2-14, B2-17, B2-23. Evidence pin: commit 3261ddd9 . Seal of record: sealed instrument manifest digest be0a11a3a4d7 . Gate-M constants are sealed preregistered design values under PM1-C004. Gate-M status: **PENDING**. The two family panels above are separate boards, stated separately and never compared with each other. Artifact links are withheld for double-blind review; withheld twins are deliberately excluded from every artifact set and are not releasable. **v2**.

B2-01, B2-02, B2-03, B2-04, B2-09, B2-10, B2-11, B2-13, B2-14, B2-17, B2-23